MIL-R-9208B(AS)
7 July 1968
SUPERSEDING
MIL-R-9208A(ASG)
1 June 1955

MILITARY SPECIFICATION

REPAIR MATERIAL, SELF-SEALING FUEL TANK

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers a repair material for mending self-sealing fuel tanks.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-225/5	Aluminum Alloy, Bar, Rod, and Wire, Rolled, Drawn or Cold Finished: 2017
TT-S-735	Standard Test Fluids: Hydrocarbon
UU-P-271	Paper, Wrapping, Waterproofed Kraft
MMM-A-260	Adhesives, Water-Resistant (For Sealing Waterproofed Paper)
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

FSC 4920

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal (Cont'd)

PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated Triple-Wall

STANDARDS

Federal

FED-STD-601	Rubber, Sampling and Testing (Test Method)
FED-STD-791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing (Test Method)

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(When requesting specifications, standards, and publications, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot (105), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials - Materials shall conform to existing specifications, where practicable, and shall be as specified herein. Materials which are not covered by existing specifications, or which are not specifically described herein, shall be of the best quality, and suitable for the purpose intended.

3.2 Design and construction - The repair material covered by this specification shall consist of a sandwich construction. The sandwich construction shall consist of a nylon barrier sandwiched in between two plies of nylon-cloth-supported gum, and cured to meet the performance requirements of this specification.

3.3 Thickness - The thickness of the material, including the entire sandwich

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

construction specified in 3.2, shall be 0.050 ±0.005 inch.

* 3.4 Form - The repair material shall be furnished in flat cut sheets, in lengths and widths as specified by the procuring activity (see 6.2).

3.5 Tensile strength - The tensile strength of the unaged (condition of material prior to Immersion test) material (without barrier) shall be not less than 1000 pounds per square inch (see 4.7.1).

3.6 Ultimate elongation - The ultimate elongation of the unaged material (without barrier) shall be not less than 300 percent (see 4.7.1).

* 3.7 Fuel resistance - After aging the material (without barrier) for 72 hours in Type III, TT-S-735, test fluid at 23 ±2° C (73.4 ±3.6° F), the tensile strength shall not decrease more than 50 percent of the original, and ultimate elongation shall not decrease more than 35 percent of the original (see 4.7.1).

* 3.8 Volume change - The volume change of the material (with barrier) after aging 72 hours in fuel conforming to TT-S-735, Type III, shall be not more than 20 percent (see 4.7.1).

3.9 Fuel contamination - The nonvolatile extractable material contributed by the repair material in contact with fuel shall not exceed 60 milligrams per 100 milliliters of contaminated fluid, and the gum residue shall not exceed 20 milligrams per 100 milliliters of contaminated fluid (see 4.7.2).

3.10 Water resistance - The tensile strength of the material (without barrier) after aging 72 hours in water at 49 ±2° C (120.2 ±3.6° F) shall not decrease more than 20 percent of the original (see 4.7.1).

3.11 Permeability - The permeability of the material (with barrier) shall be less than 0.025 fluid ounces per square foot per 24 hours (see 4.7.4).

* 3.12 Delamination - A force of not less than 3 pounds per inch shall be required to separate the rubber from the nylon film after immersion in TT-S-735, Type III, test fluid for 24 hours at a temperature of 23 ±2° C (73.4 ±3.6° F) (see 4.7.3).

3.13 Identification of product - Unless otherwise specified, material shall be marked to show the specification number, date of manufacture (month and year), and manufacturer's identification. The identification shall appear on one side of the sheet in rows of constantly recurring symbols from one end of the sheet to the other, spaced approximately 5 inches apart.

EXAMPLE: Specification MIL-R-9208B(AS) 3/68 XYZ Co. Specification MIL-R-9208B(AS) 3/68 XYZ Co.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

3.13.1 The symbols shall be clearly legible and not less than 3/8 inch high, and shall be applied by suitable means, using marking fluid that is not deleterious to the synthetic rubber. The markings shall not be obliterated by normal handling or the action of aromatic or nonaromatic gasoline. The color of the identifying symbols shall be red.

3.14 Storage condition - The repair material covered by this specification shall be protected from air circulation, sunlight, and temperatures exceeding 49° C (120° F).

3.15 Age limitation - Repair material covered by this specification which is over 12 months old from date of manufacture shall not be delivered by the material manufacturer or his licensee or agent to any Government service, airplane manufacturer, or accessory manufacturer for stocking or installation purposes. Repair material which is over 15 months old from date of manufacture shall not be delivered by a source other than the manufacturer or his agent to any Government service, airplane manufacturer, or accessory manufacturer for stocking or installation purposes.

3.16 Workmanship - The material shall be uniformly fabricated in accordance with the best commercial practice for this type of product, and free from cracks, cuts, holes, chafed spots, or other imperfections which might impair its usefulness. The material shall be free from dirt, contamination, mold release compounds, or other foreign matter. Sheets shall be evenly stacked.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of components and materials - In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification. In event of conflict, this specification shall govern.

4.2 Quality conformance inspections - Quality conformance inspections shall consist of all the examinations and tests required herein. Unless otherwise specified, inspection shall be performed in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

4.3 Samples and acceptance criteria for tests - Samples for tests shall consist of the following selected at random from the lot represented A lot shall consist of all material of the same formulation on the order offered for inspection at the same time.

One sample, approximately 200 square inches with barrier.
One sample, approximately 200 square inches without barrier.

Failure of any sample to meet any of the requirements of Section 3 shall be cause for rejection of the lot.

4.4 Samples and acceptance criteria for examinations - The sample size and acceptance criteria for examinations based on defects per hundred units inspection shall be as specified in Table I. Sampling and acceptance or rejection shall be in accordance with MIL-STD-105.

TABLE I

SAMPLE SIZE AND ACCEPTANCE CRITERIA FOR EXAMINATIONS

Inspection or Characteristic	Test Method	Inspection Level	AQL
Examination for appearance, construction, and workmanship	4.5.1	II	0.65
Examination for dimensional defects	4.5.2	II	0.65
Examination for preparation for delivery	4.5.3	II	2.5

4.5 Examinations -

4.5.1 Examination of the end product for defects in appearance, construction, and workmanship - The lot size for this examination shall be expressed in units of square feet of repair material. The sample unit shall be one square foot. Sufficient sheet stock shall be selected at random so that by examining approximately an equivalent of 20 linear feet of sheet stock the required sample footage will be obtained.

EXAMINE	DEFECTS
Check both sides of repair material	
Form	Not in sheet form

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

EXAMINE	DEFECTS
Cleanness	Not clean, evidence of mold release compound, or presence of other foreign matter.
Workmanship	Evidence of delamination or embrittlement. Any crack, cut, hole, blister, chafed spot or other defects that would impair usefulness of material.
Construction	Not uniform, ply missing.
Identification markings	Illegible, incorrect, incomplete, or omitted. Not in continuous rows lengthwise on sheet, spaced approximately 5 inches apart. Color not red, as specified.

4.5.2 Examination of the end product for dimensional defects - The sample taken in 4.5.1 shall be utilized for examination of the following defects.

EXAMINE	DEFECTS
Sheet width	More than 1/4 inch greater than specified width. Less than specified width by more than 1/4 inch.
Sheet length	More than 1/4 inch greater than specified length. Less than specified length by more than 1/4 inch.
Nominal material thickness	Not within specified tolerances.
Identification markings	Lettering less than 3/8 inch high.

4.5.3 Examination of the end product for defects in preparation for delivery - An examination shall be made to determine that packaging, packing, and markings comply with the requirements of Section 5 of this specification.

NOTE: The containers, from which the sample sheets are selected for examination under 4.5.1 and 4.5.2 shall also be used for examination under 4.5.3.

The sample unit shall be one shipping container, fully packed, selected just prior to

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

EXAMINE	DEFECTS
Packing	Not in accordance with contract requirements.
	Not packed for overseas shipment, when specified.
	Container material not as specified; closures not accomplished by specified or required methods or materials.
	No paper, or unsuitable paper, between sheets of repair material.
	Paper between sheets does not extend over full contact area between sheets.
	Inside height of container exceeds 10 inches.
Markings	Interior or exterior markings (as applicable) illegible, incorrect, omitted, or not in accordance with requirements (see 5.3).

4.6 Test conditions - The material selected for testing shall be conditioned for 24 hours at 70 to 76° F and 50 ±5 percent relative humidity and shall be tested under these conditions unless otherwise specified in the method of test. For referee purposes the conditions shall be 23 ±2° C (73.4 ±3.6° F) and 50 ±5 percent relative humidity.

4.7 Test methods -

4.7.1 Tensile strength, ultimate elongation, thickness, and immersion - The tensile strength, ultimate elongation, thickness, and immersion tests shall be performed in accordance with FED-STD-601.

4.7.2 Fuel contamination - A 5-gram sample of material (without barrier) shall be diced up into approximately 1/16-inch squares and placed in a flask containing 250 milliliters of TT-S-735, Type III, test fluid and allowed to stand for 48 hours at 23 ±2° C (73.4 ±3.6° F). The contaminated test fluid shall be decanted off and the gum residue determined by Method 3302, of FED-STD-791, except that the time of evaporation shall be 45 minutes. After cooling, the test beakers shall be weighed and the gum content determined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

The beakers shall be then placed in an appropriate solid metal block bath and maintained at 300 ±5° C (572 ±9° F) for 30 minutes. After cooling to room temperature in a closed container, the beaker shall be weighed and the gums determined. The necessary correction for preformed gums in the test fluid shall be made in each case.

4.7.3 Delamination - Three 1- by 6-inch samples (with barrier) shall be cut from the repair material. Each specimen shall be separated 1 inch along the barrier, at one end. The three specimens shall be immersed for 24 hours in TT-S-735, Type III, fluid at 23 ±2° C (73.4 ±3.6° F). Immediately after removal from the fluid, the separate ends of the specimens shall be placed in a suitable testing machine having a jaw separation of 2 inches per minute. All readings shall be the numerical average of the peak loads. All peel tests shall be conducted at a temperature of 23 ±2° C (73.4 ±3.6 F).

4.7.4 Permeability - Two disks 2-1/2 inches in diameter shall be cut from the repair material. One hundred milliliters of TT-S-735, Type III, test fluid shall be placed in a cup conforming to Figure 1. A nylon solution of 10 percent of duPont's Type FM6001 or FM6501 nylon, or equivalent, in ethyl alcohol, maintained at 49 ±2° C (120.2 ±3.6° F), shall be applied to the face of the cup flange as specified on Figure 1. When the nylon solution is almost dry, the test disk shall be applied to the cup with the barrier, if any, facing outward. The assembly shall be completed by attaching the bolting ring shown on Figure 1 and tightening the bolts from 15 to 20 inch-pounds. The cups shall be placed in a suitable rack and maintained at a temperature of 23 ±2° C (73.4 ±3.6° F) and a relative humidity of 50 ±5 percent for a 1-hour equilibration period. At the end of the 1-hour equilibration period, the cup shall be weighed to the nearest 0.005 gram and placed in the rack with the face of the cup facing upward. The cup shall be maintained at a temperature of 23 ±2° C (73.4 ±3.6 F) and a relative humidity of 50 ±5 percent for a 24-hour period. The cup shall then be weighed to check for excessive vapor loss. The bolts shall be retorqued if necessary. The cup shall be inverted (test disk down) in a rack that permits free acess of air to the test disk. The cups shall be weighed at the end of the third, fifth, and eighth day after inverting. Defective films or leaks caused by faulty assembly will usually be found when weighing on the third day. The diffusion rate calculation shall be made on the fifth- to the eighth-day period, and expressed as fluid ounces per square foot per 24 hours.

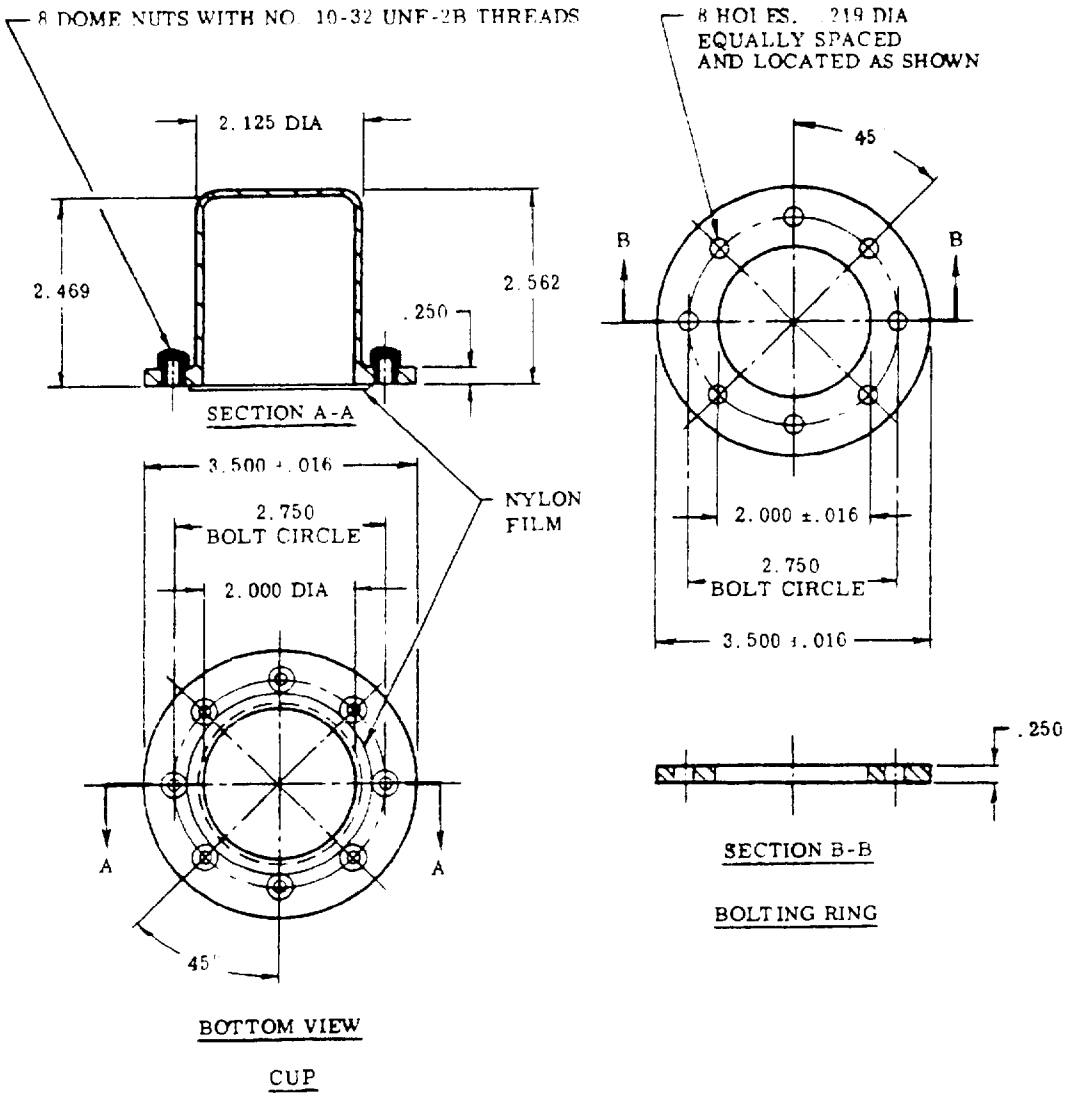
NOTE: Diffusion expressed in fluid ounces per square foot per 24 hours equals the gram loss of the test specimen per 24 hours multiplied by a factor K which is defined as follows:

$$K = \frac{144}{(\text{Specific gravity})\,(29.573)\,(3.142)\,(R)}$$

Where:

Specific gravity = Specific gravity of test fluid at 23 ±2° C (73.4 ±3.6° F)
R = Inside radius of the test cup in inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.



BREAK ALL EDGES .031 MAX RAD.
ALL FILLETS .031 MAX RAD.
MATERIAL: ALUMINUM ALLOY BAR STOCK PER QQ-A-225/5, OR EQUAL.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: ±.010.

FIGURE 1. CUP FOR PERMEABILITY TEST

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging - Not required.

5.2 Packing - Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A - Sheets, in the quantity specified by the procuring activity, shall be packed snugly, in overseas type containers conforming to PPP-B-591, PPP-B-601, or PPP-B-621. Containers shall be furnished with sealed case liners made from material conforming to UU-P-271. All seams and closures shall be sealed with adhesive conforming to MMM-A-260. The sheets shall be interleaved with any suitable paper extending over the full area of contact between the sheets. The inside height of the container shall not exceed ten inches.

5.2.2 Level B - Sheets shall be packed for shipment in fiberboard boxes conforming to PPP-P-636, Class - Domestic, or PPP-B-640, Class 1. Closure and sealing shall be in accordance with the appendix of the applicable specification.

5.2.3 Level C - Sheets shall be packed in a manner that will insure carrier acceptance and safe delivery to first destination for immediate use. This maybe the supplier's commercial practice when it meets these requirements.

5.3 Marking of shipments - Containers shall be marked in accordance with MIL-STD-129 and as follows:

REPAIR MATERIAL, SELF-SEALING FUEL TANK
Specification MIL-R-9208B(AS)
Length 1/
Width 1/
Stock No. 1/ (Federal, or Navy, as applicable)

1/ Applicable data to be entered by the contractor. If the Federal Stock No. is available (either stated in the purchase document or furnished by the procuring activity), omit the Navy Stock No. If the Federal Stock No. is not available, insert the Navy Stock No., and leave a space for subsequent placement of the Federal Stock No.

6. NOTES

6.1 Intended use - The repair material covered by this specification is intended for use in the repair of self-sealing fuel tanks conforming to MIL-T-5578.

* 6.2 Ordering data - Procurement documents should specify:

(a) Title, number, and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

(b) Length, width, and quantity of sheets (see 3.4).

(c) Applicable levels of packaging and packing required (see Section 5).

NOTE: Repair material should not be ordered for more than 3 months' use.

* 6.3 Changes from previous issue - The outside margins of this document have been marked "*" to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:04Z
Check the source to verify that this is the current version before use.